

INFIBEAM AVENUES

About the company:

Infibeam Avenues Limited, founded in 2007 and headquartered in GIFT City, Gandhinagar, is a leading Indian fintech company. It operates the popular payment gateway **CCAvenue** and offers digital payment and e-commerce technology solutions to businesses, governments, and financial institutions. The company serves clients across India and internationally, with presence in the Middle East, USA, and Australia. Listed on the NSE and BSE, Infibeam Avenues processes high transaction volumes daily and focuses on innovations in digital payments, data security, and enterprise SaaS platforms, supporting India's growing digital economy.

Online Assessment:

- **Section 1** comprised 20 multiple-choice questions covering pseudocode output prediction, data structures and algorithms concepts, and computer fundamentals. It tested candidates' understanding of logic flow, problem-solving ability, and core topics such as operating systems, networking, and DBMS, evaluating overall technical aptitude and analytical thinking skills.
- **Section 2** contains two DSA questions.
 - **Problem 1:** Calculate height of a binary tree but neglect if node's value is odd.
 - **Problem 2:**
Input: Two integer inputs will be given. One integer says the value and other integer says in how many digits the value should be expressed.
Output: String
 - Example 1:
Input: Value: 999, Digits: 3
Output: "999"
 - Example 2:
Input: Value: 1000, Digits: 3
Output: "A00"

Here A means 10, B means 11, ... goes till Z.

[**Hint for Section2:** Write logic/approach for the code as comments before start coding and focus to score on Section 1]

I was able to solve only 1st problem half the testcases provided. I didn't solve the second one.

13 candidates were shortlisted for the interview.

Interview (Offline and Virtual mixed at CUIC):

All three rounds are technical. [No HR or managerial rounds]

- **Round 1 - Offline:**

The interview began with a self-introduction, followed by two Data Structures and Algorithms (DSA) problems and a puzzle.

- **Problem 1:** A modified linked-list problem where each node had both a child and a next pointer, resembling a tree structure. The task was to flatten this structure into a linear linked list by traversing through child nodes first, then next nodes. I initially solved it using a stack-based iterative DFS approach. Upon being asked to optimize space complexity, I proposed a recursive DFS solution. The interviewer later requested pseudocode, which I couldn't provide at that moment.
- **Problem 2:** Given a binary tree, the objective was to find the sum of the maximum possible Binary Search Tree (BST) within it. Each leaf node was considered a BST, and the goal was to compute the highest sum across all possible BSTs. Only the logic was required, and the interviewer provided guidance during the discussion.
- **Puzzle:** [Sand Timer Puzzle with solution | PrepInsta](#). I was unable to solve it during the interview.

4 out of 13 candidates were shortlisted for next round.

- **Round 2 – Virtual:**

I began with a self-introduction, followed by one puzzle and two data structures and algorithms (DSA) problems.

- **Puzzle:** [Puzzle | The Burning Candles - GeeksforGeeks](#). I initially struggled to solve it on my first attempt, but with guidance from the interviewer, I was able to arrive at the correct solution.
- **Problem 1:** [Jump Game - LeetCode](#). I started with an $O(n^2)$ approach and then optimized it to an $O(n)$ solution. I was asked to implement the code in an online compiler.
- **Problem 2:** [Gas Station - LeetCode](#). I was required to explain the logic without coding.

3 out of 13 candidates were shortlisted for next round.

- **Round 3 – Virtual:**

This round began with a self-introduction, followed by a DSA problem shared through Codeshare, where code execution was not allowed.

- **Problem:** Given two sorted integer arrays, return the range $[a, b]$ such that $|a - b|$ is minimal, where a and b represent one element from each array.
- **Approach:** I initially proposed an $O(n \log n)$ solution and, with guidance from the interviewer, optimized it to an $O(n)$ two-pointer approach.

At the end of this round, **two candidates were offered positions.**